Linear Algebra — numbering: 10/21 version → new version (Chapters 1–8)
編號對照表：10/21 版 → 新版（第 1–8 章）

新版調整了部分項目的編號；箭頭左邊是 10/21 版的編號，右邊是新版的項目，灰色的項目編號不變。
標為「moved to the regression chapter」的項目，會收入稍後發布的迴歸一章。
標為「moved to Chapter 13」的項目，會收入稍後發布的新版第 13 章。
Left of the arrow: the number in the 10/21 version; right: the item in the new version. Grey: the same number in both versions.

Chapter 1. Matrices

Main text

1.1 to 1.11: same numbers
1.12 → Definition 1.14
1.13 → Proposition 1.15
1.14 → Example 1.16
1.15 → Example 1.17

Exercises

1.1 → Exercise 1.4
1.2 → Exercise 1.3
1.3 → Exercise 1.10 [only (b)]
1.4 → Exercise 1.5
1.5 → Exercise 1.2
1.6 → Exercise 1.14
1.7 → Exercise 1.12
1.8 → Exercise 1.6
1.9 → Exercise 1.17 [(b) changed]
1.10 → Exercise 1.18
1.11 → Exercise 1.7
1.12 → Exercise 1.8

Problems

1.1 to 1.2: same numbers
1.3 → Problem 1.5 [only (b); (a) is in the solution]
1.4 → Problem 1.3
1.5 → Problem 1.4
1.6 same number
1.7 → *merged into* Corollary 1.31 and Problem 1.17
1.8 → *deleted:* duplicate of Proposition 1.21; see also Problem 2.3
1.9 → Problem 1.7
1.10 → *deleted:* duplicate of Proposition 1.29(b),(d) and Problem 6.4(a)
1.11 → *deleted:* partly Proposition 1.33; see also Example 6.27

Chapter 2. Gauss elimination, RREF, and inverses

Main text

2.1 to 2.7: same numbers
2.8 → Theorem 2.8 [was a Proposition]
2.9 to 2.12: same numbers
2.13 → Proposition 2.18
2.14 → Example 2.19
2.15 → Theorem 2.13
2.16 → Definition 2.14
2.17 → Theorem 2.15
2.18 → Theorem 2.16
2.19 → Example 2.20
2.20 → Example 2.21
2.21 → Example 2.22
2.22 → Theorem 2.26
2.23 → Theorem 2.27
2.24 → Example 2.28
2.25 → Example 2.29
2.26 → Example 2.30

Exercises

2.1 → Exercise 2.10
2.2 → Exercise 2.6
2.3 → Exercise 2.19
2.4 → Exercise 2.8 [other data]
2.5 → Exercise 2.9
2.6 → Exercise 2.12 [other data]
2.7 → Exercise 2.13
2.8 → Exercise 2.14
2.9 → Exercise 2.3
2.10 → Exercise 2.4
2.11 → Exercise 2.23

Problems

2.1 to 2.3: same numbers
2.4 → *deleted:* covered by Corollary 2.24 and Theorem 2.3(b)
2.5 → Problem 2.4 [new statement]
2.6 → *deleted:* (a) is Corollary 2.23(a)
2.7 → *deleted:* (a), (b) are Theorem 5.32(b)
2.8 → *deleted:* (a), (b), (d) are Theorem 2.31; (c) is a case of Problem 5.6(b)
2.9 → *deleted:* LU factorization; see Proposition 8.38
2.10 → *deleted:* numerical; uses Problem 2.9 above

Chapter 3. Vector spaces, subspaces, span, and bases

Main text

3.1 to 3.9: same numbers
3.10 → Definition 3.11
3.11 → Proposition 3.12
3.12 → Proposition 3.13
3.13 → Example 3.15
3.14 → Definition 3.17
3.15 → Theorem 3.18
3.16 → Example 3.19 [other data]
3.17 → Example 3.20
3.18 → Example 3.21
3.19 → Example 3.22
3.20 → Definition 3.30
3.21 → Theorem 3.31
3.22 → Theorem 3.32
3.23 → Theorem 3.33
3.24 → Example 3.34
3.25 → Example 3.35

Exercises

3.1 → Problem 3.1 [was an Exercise]
3.2 → Problem 3.2 [was an Exercise; only (a), with 3×3 matrices]
3.3 → Exercise 3.2
3.4 → Exercise 3.6 [other data]
3.5 → Exercise 3.3
3.6 → Exercise 3.14
3.7 → Exercise 3.9 [other data]
3.8 → Exercise 3.7
3.9 → Exercise 3.11 [other data]
3.10 same number
3.11 → Exercise 3.13
3.12 → Exercise 3.18 [other data]
3.13 → *deleted:* a similar drill is Exercise 3.19

Problems

3.1 → Problem 3.4
3.2 → Problem 3.3 [new statement]
3.3 → *deleted:* duplicate of Proposition 3.14(b) and Proposition 3.13(c)
3.4 → *deleted:* no book source; see Theorem 3.10(b) and Theorem 3.28
3.5 → *deleted:* no book source; see Theorem 2.31(a) and Proposition 3.12
3.6 → *deleted:* no book source; see Theorem 4.3(a)
3.7 → *deleted:* no book source; see Example 3.22
3.8 → Problem 3.5 [without the numerical example]
3.9 → Problem 3.6 [new statement]
3.10 → *merged into* Problem 3.6(b)

Chapter 4. Bases, dimension, and rank

Main text

4.1 to 4.22: same numbers

Exercises

4.1 → Exercise 4.2
4.2 → Exercise 4.3
4.3 → Exercise 4.1
4.4 → Exercise 4.8 [other data]
4.5 → Exercise 4.4
4.6 same number
4.7 → Exercise 4.10
4.8 → Exercise 4.15 [other data]
4.9 → Exercise 4.16 [other data]
4.10 → Exercise 4.13

Problems

4.1 → Problem 4.4
4.2 → *deleted:* follows from Theorem 4.21(a) and Theorem 4.15(b)
4.3 → *deleted:* duplicate of Proposition 4.26 and Theorem 4.5(a)
4.4 → Problem 4.2
4.5 → *deleted:* special case of Theorem 4.35(b)
4.6 → Problem 4.1 [new statement]
4.7 → Problem 4.3
4.8 → *deleted:* follows from Theorem 6.29

4.9 → Problem 5.9 [new statement]
4.10 → Problem 4.5 [new statement]

Chapter 5. Block matrices and determinants

Main text
5.1 to 5.11: same numbers
5.12 → Definition 5.13 [new definition]
5.13 → Theorem 5.21
5.14 → Example 5.23
5.15 → *merged into* Theorem 5.14(c) and Theorem 5.22
5.16 → *merged into* Definition 5.13, Theorem 5.14 and Theorem 5.22
5.17 → Example 5.16
5.18 → Example 5.17
5.19 → Theorem 5.18
5.20 → Theorem 5.24
5.21 → Theorem 5.27
5.22 → Remark 5.25
5.23 → Example 5.26
5.24 → Example 5.28
5.25 → Theorem 5.29
5.26 → Theorem 5.30
5.27 → Theorem 5.31
5.28 → Example 5.33
5.29 → Theorem 5.34
5.30 → Example 5.35

Exercises
5.1 → Exercise 5.2
5.2 → Exercise 5.3
5.3 → Exercise 5.4
5.4 → Exercise 5.8
5.5 → Exercise 5.12
5.6 → Exercise 5.14 [other data]
5.7 → Exercise 5.16
5.8 → Exercise 5.10
5.9 → Exercise 5.18
5.10 → Exercise 5.20
5.11 → Exercise 5.22
5.12 → Exercise 5.19

Problems
5.1 to 5.3: same numbers
5.4 → *deleted:* special case B = 0 of Theorem 5.12(a)
5.5 → Theorem 5.12 [was a Problem]
5.6 → Problem 5.5
5.7 → Problem 5.6
5.8 → Problem 5.7
5.9 → *merged into* Example 5.33
5.10 → Problem 5.8 [without the counterexample]

Chapter 6. Orthogonality and projections

Main text
6.1 to 6.2: same numbers
6.3 → Theorem 6.4
6.4 → Definition 6.5
6.5 → Theorem 6.6
6.6 → Corollary 6.7
6.7 → Example 6.10
6.8 → Proposition 6.11
6.9 → Definition 6.12
6.10 → Proposition 6.13
6.11 → Theorem 6.14
6.12 → Theorem 6.16
6.13 → Example 6.17
6.14 → Definition 6.22
6.15 → Theorem 6.23
6.16 → Theorem 6.29
6.17 → Theorem 6.8(a) [was a Proposition]
6.18 → *merged into* Theorem 6.31(a) and Definition 6.5
6.19 → Theorem 6.31 [parts changed: old (b) is (c), old (c) is Theorem 6.34]
6.20 → Corollary 6.32
6.21 → Corollary 6.35
6.22 → Example 6.27
6.23 → Example 6.28 [same data, other method]
6.24 → Corollary 6.37 [was a Theorem]
6.25 → Theorem 6.36
6.26 → *moved to the regression chapter*

Exercises
6.1 → Exercise 6.6 [other data in (a)]
6.2 → Exercise 6.3
6.3 → Exercise 6.2
6.4 same number
6.5 → Exercise 6.14
6.6 → Exercise 6.15
6.7 → Exercise 6.16
6.8 → Exercise 6.13
6.9 → *moved to the regression chapter*
6.10 → *moved to the regression chapter*

Problems
6.1 → Example 6.21(a),(b) [was a Problem]
6.2 → *deleted:* no suitable book source
6.3 → *moved to the regression chapter*
6.4 → *deleted:* duplicate of Problem 6.1(b) and Theorem 6.8(b)
6.5 → *deleted:* no suitable book source
6.6 → *deleted:* see Problem 6.4 and Corollary 6.37
6.7 → Problem 6.6 [only (a)]
6.8 → *deleted:* no suitable book source
6.9 → Problem 6.3 [only (a)]
6.10 → *deleted:* no suitable book source
6.11 → *deleted:* no suitable book source

Chapter 7. Eigenvalues and diagonalization

Main text
7.1 → Definition 7.1 [geometric multiplicity: Definition 7.21]
7.2 same number
7.3 → Theorem 7.3 [(iii) and (iv) interchanged; geometric multiplicity: Definition 7.21]
7.4 → *merged into* Definition 7.4, Proposition 7.5 and Theorem 7.9
7.5 → Definition 7.7
7.6 same number
7.7 → *merged into* Definition 7.7 and Proposition 7.18(d)
7.8 → Definition 7.10 [new definition]
7.9 → *merged into* Definition 7.10
7.10 → Example 7.17
7.11 → *deleted:* (a) is Proposition 7.16(b), (b) is in the proof of Theorem 7.25; for (c), (d) see Example 7.8 and Example 7.12
7.12 → *merged into* Proposition 7.18(a),(b) and Theorem 7.19
7.13 → Definition 7.14
7.14 → Theorem 7.15
7.15 → Theorem 7.33 [was a Proposition; parts changed: old (b) is (b), (c)]
7.16 → Example 7.35
7.17 → Example 8.16 [other method]
7.18 → Definition 7.23
7.19 → *merged into* Definition 7.23
7.20 → *merged into* Lemma 7.24(b) and Theorem 7.20
7.21 → Theorem 7.22
7.22 → Theorem 7.25 [new statement]
7.23 → Example 7.28 [only (c); (a) is in the text after Theorem 7.25]
7.24 → Proposition 7.27(a)
7.25 → Theorem 8.13(a) [tr A = r is Theorem 6.29(c)]
7.26 → Example 8.14
7.27 → Corollary 6.15(b),(c) [was a Lemma]
7.28 → Theorem 7.30
7.29 → Example 7.31
7.30 → Corollary 7.32 [parts changed: old (a) is Theorem 7.19, old (b), (c) are (a), (b)]

Exercises
7.1 → Exercise 7.2
7.2 → Exercise 7.18
7.3 → Exercise 7.12
7.4 → Exercise 7.10
7.5 → Exercise 7.13
7.6 → Exercise 7.7
7.7 → Exercise 7.15
7.8 → Exercise 7.16

Problems
7.1 → *deleted:* no suitable book source; see Problem 7.7
7.2 → Problem 7.14 [only (a), (b)]
7.3 → Problem 7.8 [new statement]
7.4 → *deleted:* see Example 8.16(b),(c)
7.5 → Problem 7.9 [(a) changed; (c) is Example 7.40]
7.6 → Problem 7.6 [only (b) and the first part of (a); (c) is Problem 7.5(c)]

7.7 → *deleted:* no suitable book source; see Theorem 6.29(c) and Problem 5.7

Chapter 8. The spectral theorem and quadratic forms

Main text

8.1 → *deleted:* Theorem 8.3 is proved without calculus
8.2 → *deleted:* Theorem 8.3 is proved without it; for the largest value of the quadratic form on the unit sphere see Theorem 8.21(d)
8.3 → *deleted:* an example of Lemma 8.2 above
8.4 → Theorem 8.3 [parts changed: old (a) is (a) and (b), old (b) is (c) and Theorem 8.4, old (c) is Theorem 8.2, old (d) is in (b)]
8.5 → *merged into* Theorem 8.4 and Theorem 8.6
8.6 → Example 8.8
8.7 → Definition 8.9
8.8 → Proposition 8.10
8.9 → Example 8.12
8.10 → *merged into* Theorem 6.31(b),(c) and Theorem 8.13(c)
8.11 → Definition 8.20 [quadratic forms: Definition 8.17]
8.12 → *deleted:* 2 × 2 examples; for (b) see Example 8.19(b)
8.13 → Theorem 8.21 [parts changed: old (a) and (b) are (a), old (c) is (d), old (d) is (b)]
8.14 → Definition 8.29
8.15 → *merged into* Theorem 8.21(c), Proposition 8.23(c) and Theorem 8.30
8.16 → *merged into* Theorem 8.24(a),(b), Theorem 8.25 and Theorem 8.26(a)
8.17 → *merged into* Corollary 8.27 and Theorem 8.30
8.18 → Theorem 8.31 [was a Lemma]
8.19 → Theorem 8.34
8.20 → *deleted:* the example is the 2 × 2 case of Problem 8.4
8.21 → Example 8.36
8.22 → Theorem 8.40 [new statement: the factor is upper triangular]
8.23 → Remark 8.41
8.24 → Example 8.42
8.25 → Theorem 8.46
8.26 → Example 8.47
8.27 → Definition 8.48
8.28 → Proposition 8.50 [parts changed: old (c) is (d), old (d) is (c) and Theorem 8.51, old (e) is Theorem 8.52]
8.29 → Example 8.49
8.30 → *merged into* Theorem 8.57 and Corollary 8.58(b)
8.31 → Example 8.59
8.32 → *moved to* Chapter 13
8.33 → *moved to* Chapter 13
8.34 → *moved to* Chapter 13
8.35 → *moved to* Chapter 13
8.36 → *moved to* Chapter 13
8.37 → *moved to* Chapter 13

Exercises

8.1 → Exercise 8.4
8.2 same number
8.3 → Exercise 8.6
8.4 → Exercise 8.9 [other data in (a)–(c)]
8.5 → Exercise 8.12
8.6 → Exercise 8.14
8.7 → Exercise 8.17
8.8 → Exercise 8.21
8.9 → *moved to* Chapter 13
8.10 → *moved to* Chapter 13
8.11 → *moved to* Chapter 13

Problems

8.1 → Problem 8.7
8.2 → Problem 8.11 [only (a)]
8.3 → *deleted:* (a), (b) are in Example 8.16(c) and Example 8.22(c); (c) moved to Chapter 13
8.4 → Problem 8.5 [only (b), for the rows; (a) is Theorem 8.32]
8.5 → *deleted:* no suitable book source; for (b) see Theorem 8.30
8.6 → *deleted:* (a), (b) are in a theorem of a later chapter; (c) moved to Chapter 13
8.7 → *deleted:* no suitable book source
8.8 → *moved to* Chapter 13